UNIT 4 · TOPIC 7(B)

# Recent Trends in Tractor Design

*Syllabus topic: Recent trends in tractor design; emissions and control of pollutants*

## Powertrain and hydraulics

- CRDI electronic diesels; turbocharging as standard; TREM-V after-treatment (Topic 7a).
- Transmissions: synchromesh → powershift/shuttle → CVT-IVT on premium models.
- Electro-hydraulic hitch control (EHC); load-sensing hydraulics.

## Engine technology shift under TREM norms (as of July 2026)

Mechanical inline/rotary FIPs are giving way to **electronic common-rail (CRDI)** injection on Indian tractors as TREM-IV (>50 hp, since Jan 2023) and TREM-V (56–560 kW & 8–19 kW from 1 Oct 2026) tighten limits:

- Rail pressures **1200–2000 bar**; solenoid/piezo injectors; multiple injection events.
- ECU governing — an electronic governor replaces the flyweights.
- **Service skill shift:** diagnostics move from feeler gauge + nozzle tester to OBD scan tools — but base mechanical adjustments (valve clearance, belt tension, bleeding) remain on the vast <50 hp fleet.

## Electrification, alternative fuels and autonomy

- Commercial e-tractors on sale: e.g., **Sonalika Tiger Electric**; **CSIR-CMERI PRIMA ET11** (indigenous design).
- Hybrid and hydrogen ($H_2$) tractor prototypes under global development.
- CBG under SATAT; India's first CNG tractor conversion unveiled 2021.
- GPS/RTK auto-steer, telematics fleet monitoring, ISOBUS implements (Unit 2, Topic 10; Unit 10).
- Battery basics: Unit 3, Topic 12c. Renewable fuels: Unit 7.

## Indian market trends

Rising 4WD and >50 hp share, AC cabins/ROPS, precision packages; India is the world's largest tractor market by volume.

## Operator-centred design

ROPS + seat belt, sound limits at the operator's ear, vibration-isolated platforms, ergonomic controls (Unit 5).

## Practice MCQs

**Q202. India's indigenous CSIR e-tractor is the:**

(a) PRIMA ET11 (b) Bharat 100
(c) Kisan-e (d) Vidyut 5

**Answers**

Q202 — (a) PRIMA ET11

**Sources:** Liljedahl, Turnquist, Smith & Hoki — Tractors and their Power Units; standard thermodynamics texts; PIB/MoPNG releases on ethanol blending; MoRTH/CMVR TREM notifications and DieselNet (as verified in the source notes, July 2026); NPTEL lectures. Liljedahl et al. — Tractors and their Power Units; Jagdishwar Sahay; manufacturer service practice; TREM notifications; NPTEL farm power lectures.

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